

1. Setup Technitium

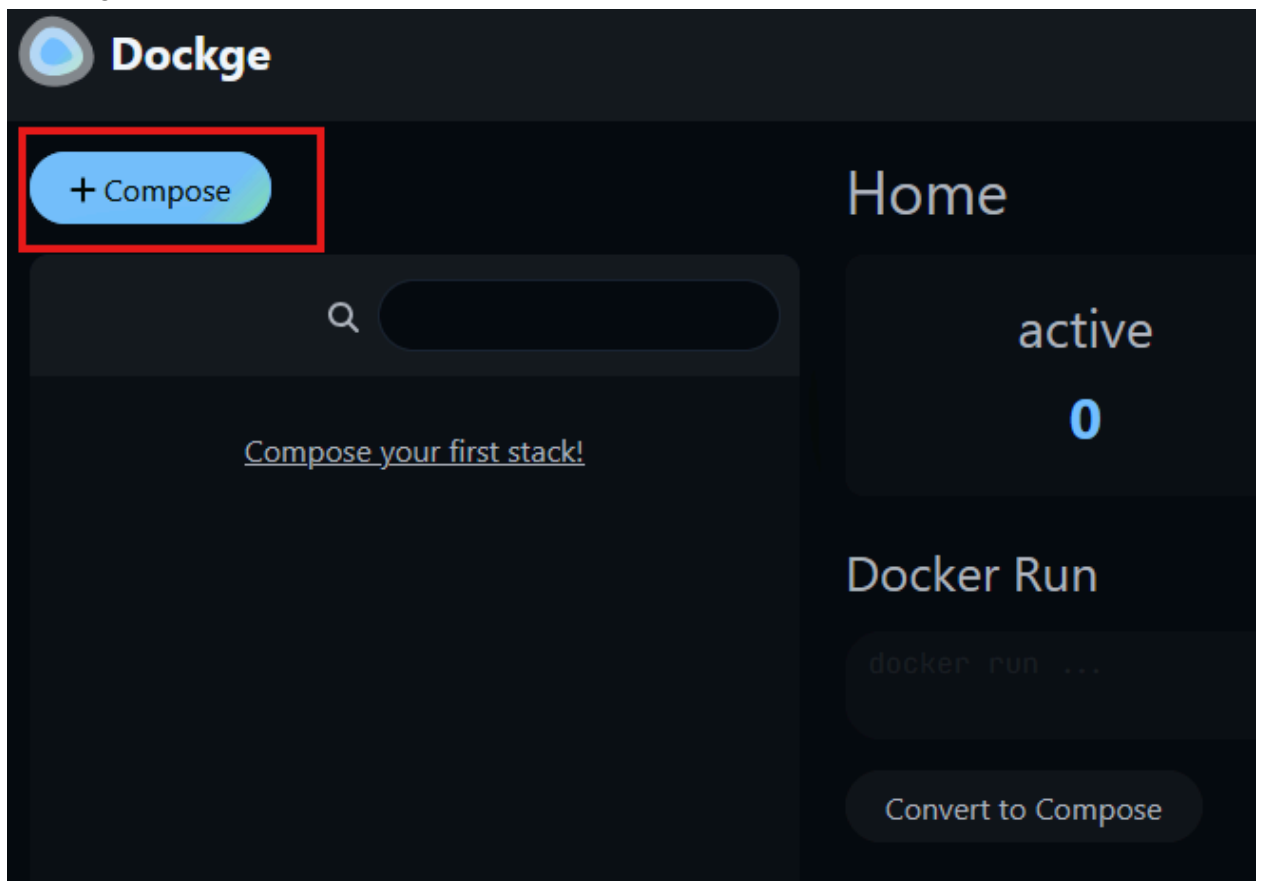
1.1 Run Technitium docker compose

Original docker compose example from Technitium:

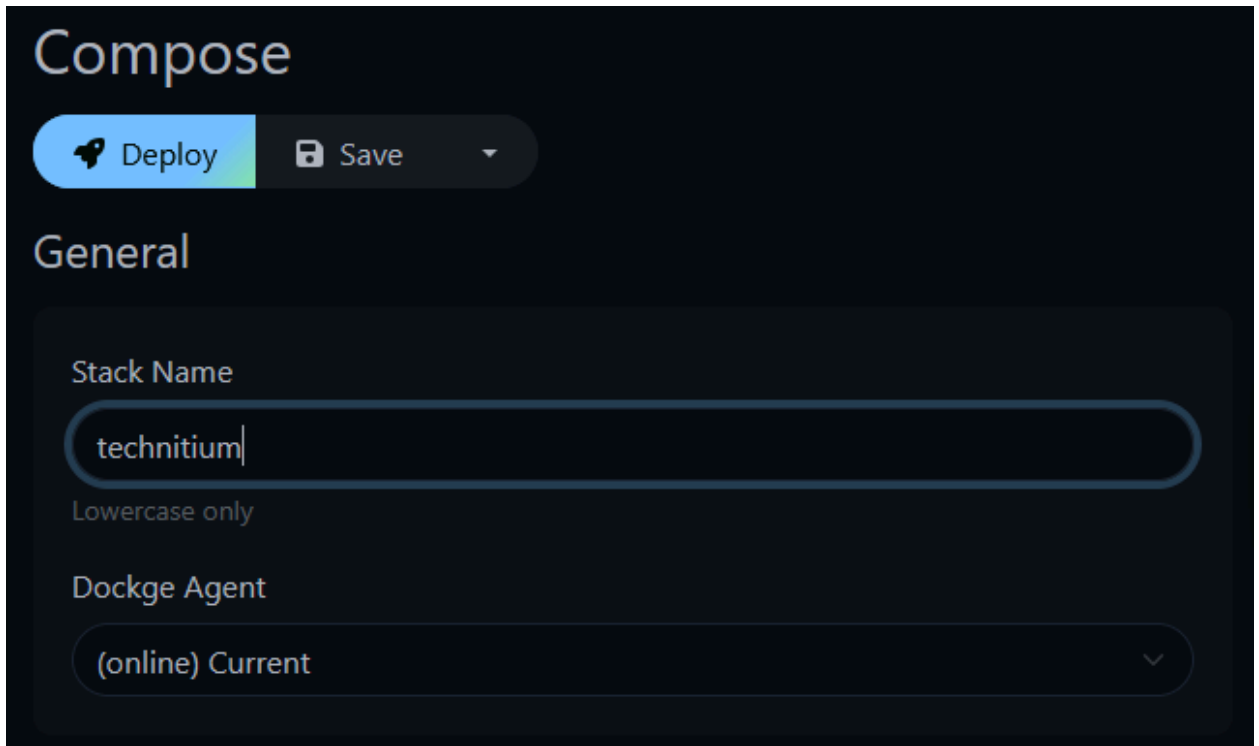
<https://github.com/TechnitiumSoftware/DnsServer/blob/master/docker-compose.yml>

For simplicity, I've slightly modified the compose file for easy deployment, and to facilitate log ingestion with Alloy in Part 3 - Log monitoring.

1. Copy the content of the compose.yaml file found here:
<https://github.com/davetjh/div0-homelab-workshop/blob/main/technitium/compose.yaml>
2. In dockge, click the +compose button in the top left corner



3. Give the stack a name, e.g. **technitium**



Compose

Deploy Save

General

Stack Name

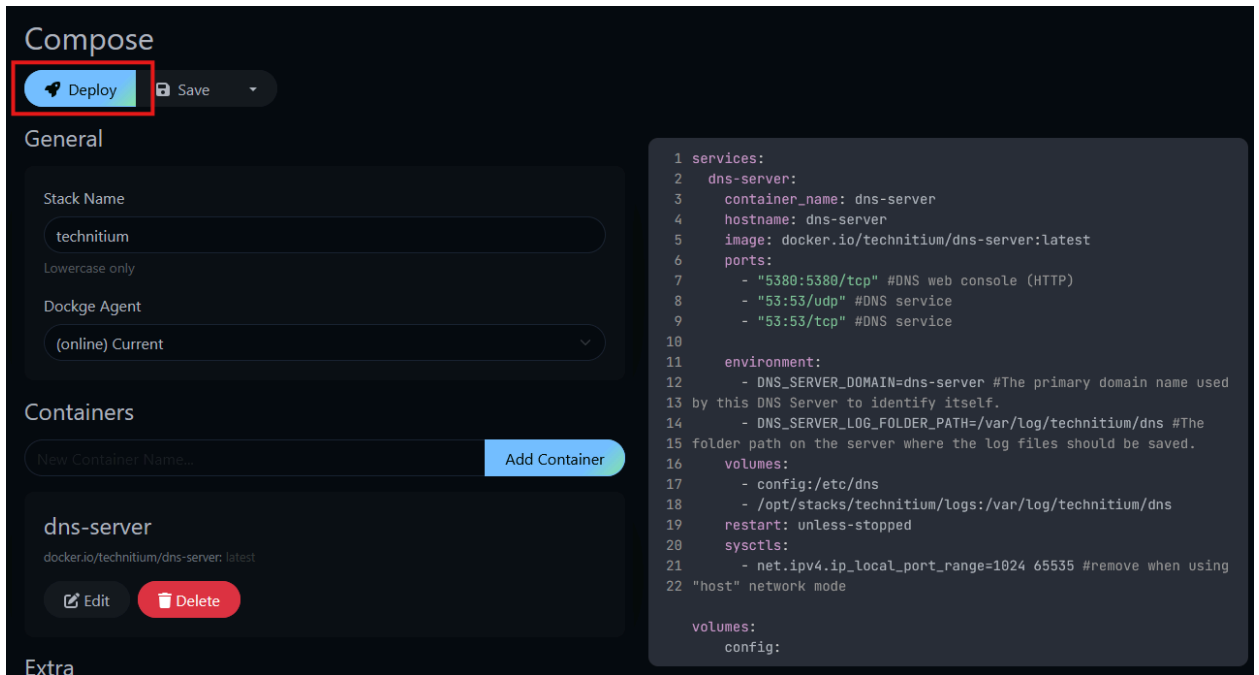
technitium

Lowercase only

Dockge Agent

(online) Current

4. In the box on the right, replace the existing content with the compose file that you just copied in step 1.
5. Click Deploy



Compose

Deploy Save

General

Stack Name

technitium

Lowercase only

Dockge Agent

(online) Current

Containers

New Container Name... Add Container

dns-server

docker.io/technitium/dns-server:latest

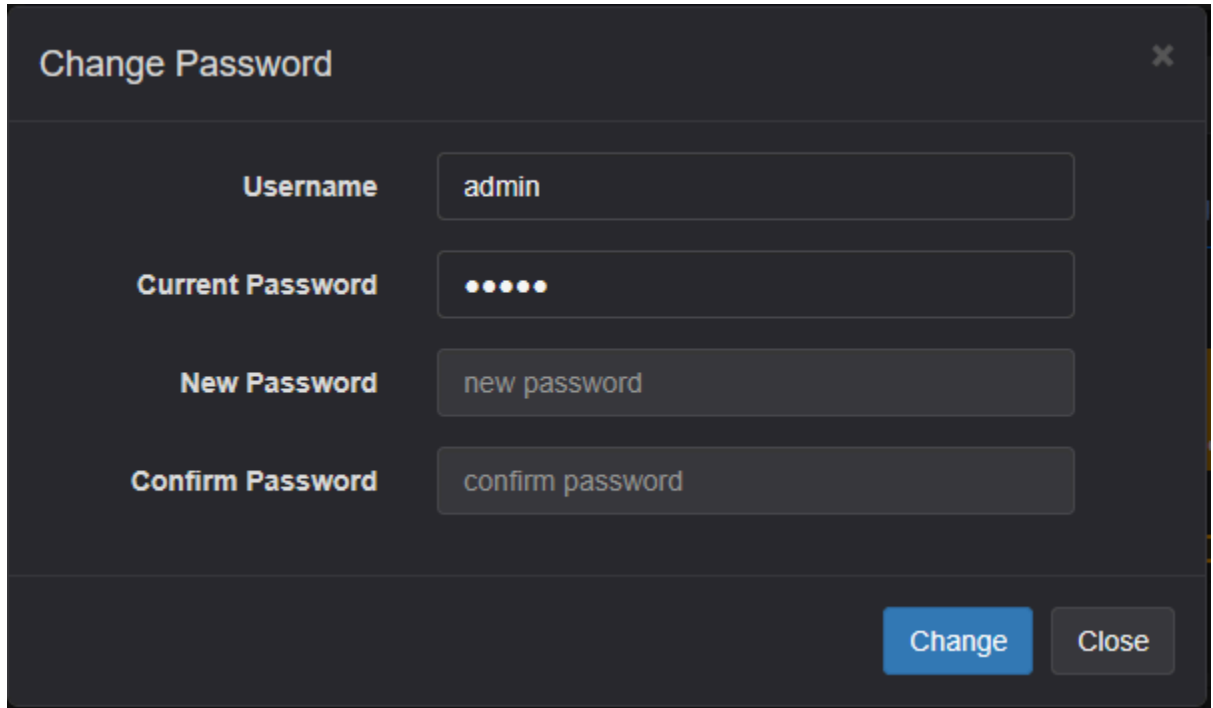
Edit Delete

Extra

```
1 services:
2   dns-server:
3     container_name: dns-server
4     hostname: dns-server
5     image: docker.io/technitium/dns-server:latest
6     ports:
7       - "5380:5380/tcp" #DNS web console (HTTP)
8       - "53:53/udp" #DNS service
9       - "53:53/tcp" #DNS service
10
11   environment:
12     - DNS_SERVER_DOMAIN=dns-server #The primary domain name used
13     by this DNS Server to identify itself.
14     - DNS_SERVER_LOG_FOLDER_PATH=/var/log/technitium/dns #The
15     folder path on the server where the log files should be saved.
16   volumes:
17     - config:/etc/dns
18     - /opt/stacks/technitium/logs:/var/log/technitium/dns
19   restart: unless-stopped
20   sysctls:
21     - net.ipv4.ip_local_port_range=1024 65535 #remove when using
22     "host" network mode
23
24   volumes:
25     config:
```

1.2 Access the Technitium portal

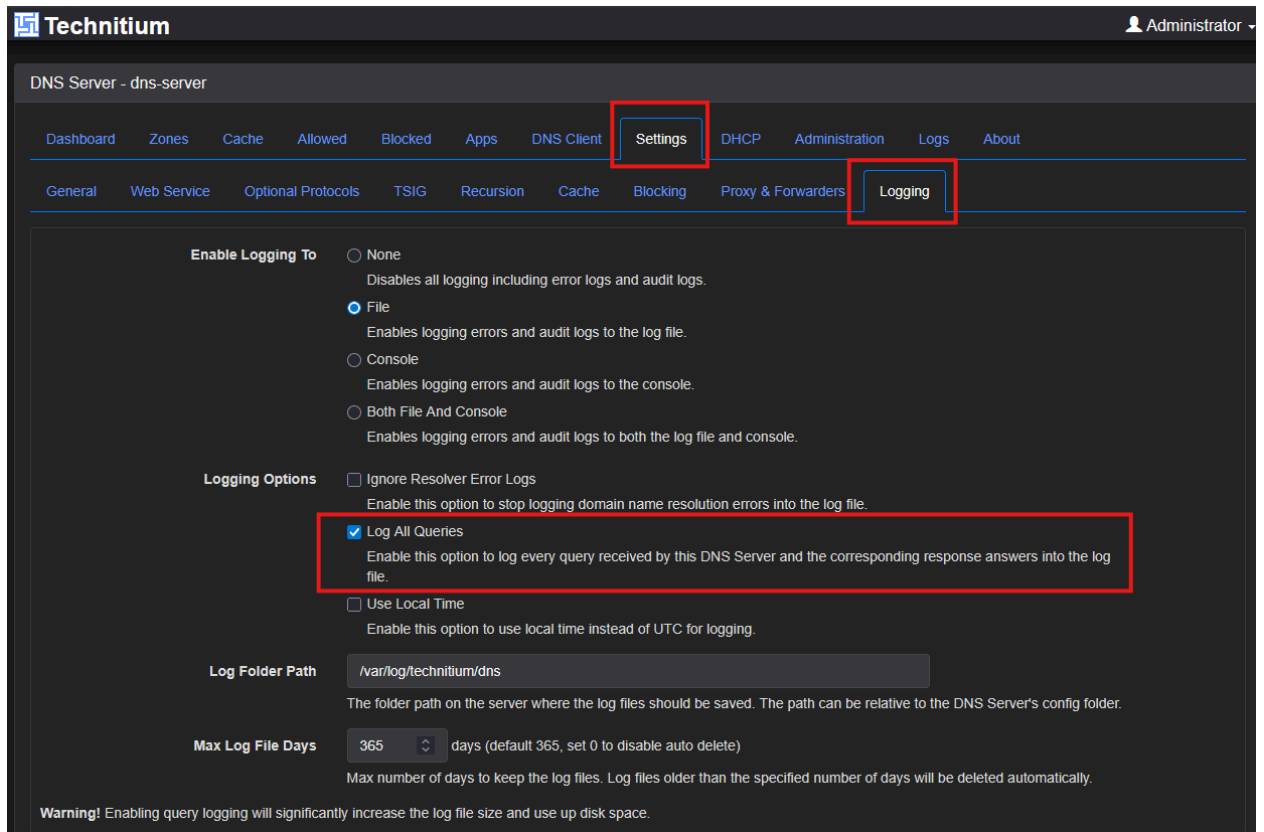
1. Access the technitium dashboard by visiting <http://<lxc-ip-addr>:5380>. You will be prompted to change the admin password



The screenshot shows a dark-themed 'Change Password' dialog box with a close button (X) in the top right corner. It contains four input fields: 'Username' with the value 'admin', 'Current Password' with five dots, 'New Password' with the value 'new password', and 'Confirm Password' with the value 'confirm password'. At the bottom right, there are two buttons: a blue 'Change' button and a grey 'Close' button.

Field	Value
Username	admin
Current Password	•••••
New Password	new password
Confirm Password	confirm password

2. To ensure that DNS query logs are recorded, go to **Settings > Logging** and enable **Log All Queries**



3. Click **Save Settings**

1.3 Configure Technitium

1. Go to settings, and then "Proxy & Forwarders"

Technitium

Administrator

DNS Server - technitiumdns

1

DashboardZonesCacheAllowedBlockedAppsDNS ClientSettingsDHCPAdministration

LogsAbout

GeneralWeb ServiceOptional ProtocolsTSIGRecursionCacheBlockingProxy & Forwarders2

Logging

Network Proxy

☒ No Proxy (default)

☐ HTTP Proxy

☐ SOCKS5 Proxy

Proxy Server Address

domain name or IP address

Proxy Server Port

port

Username

username

Password

password

Proxy Bypass List

Enter IP addresses, network addresses or domain names to never proxy.

Note! When proxy server is configured, DNS Server will use it for all outbound network requests.

Forwarders

3

Quick Select

Enter forwarder DNS Server IP addresses or URLs one below another in above text field or use the Quick Select list to select desired forwarder.

Forwarder Protocol

☒ DNS-over-UDP (default)

☐ DNS-over-TCP

☐ DNS-over-TLS

☐ DNS-over-HTTPS

☐ DNS-over-QUIC

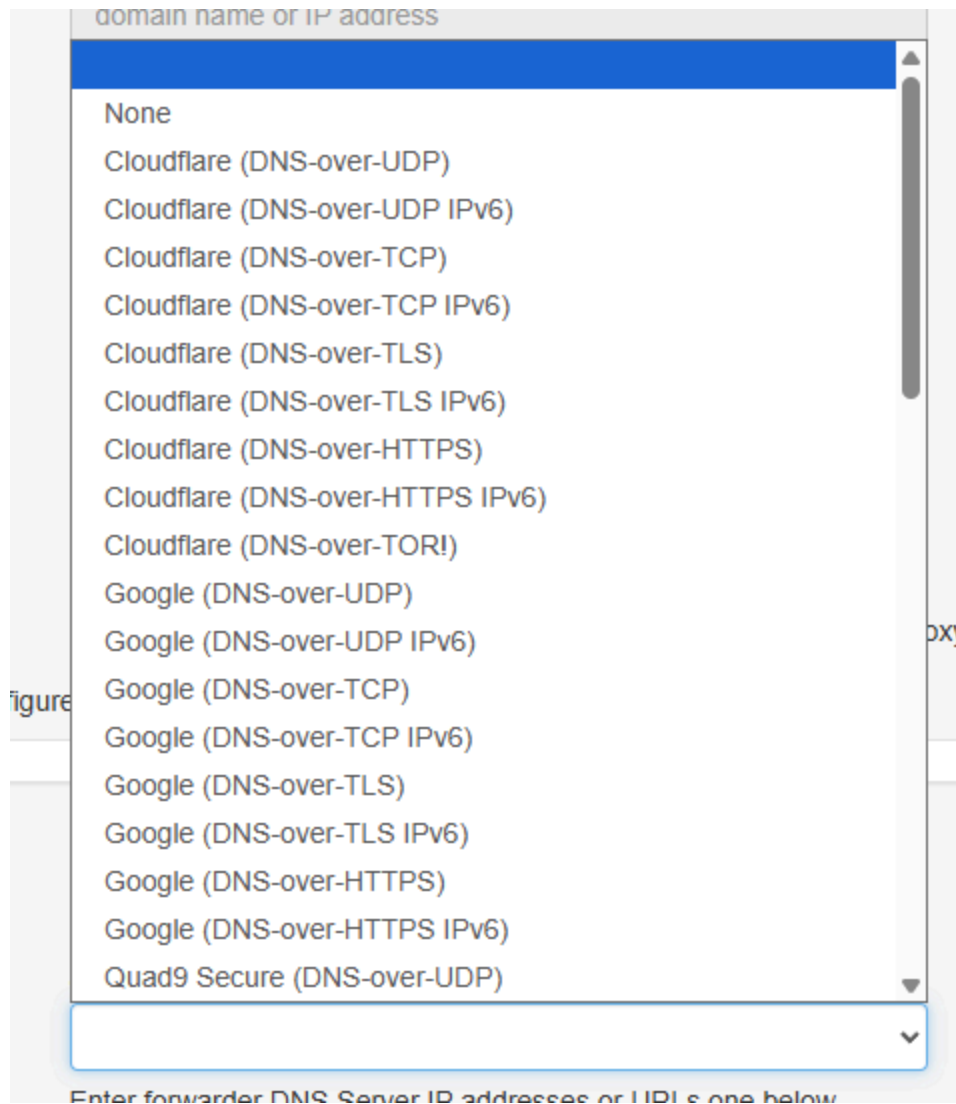
Select a protocol that this DNS Server must use to query the forwarders specified above.

Concurrent Forwarding

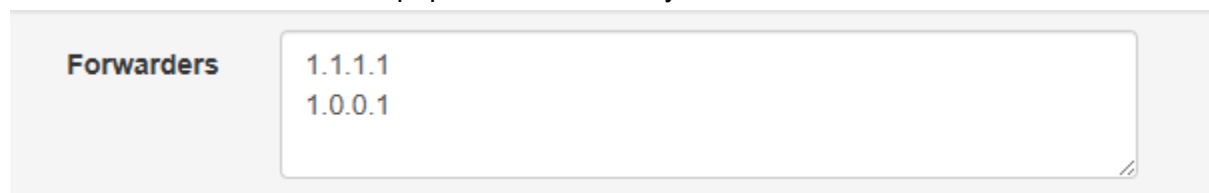
☒ Enable Concurrent Forwarding

Enable this option to allow querying two or more forwarders concurrently instead of sequentially querying them in their given order. The DNS Server will automatically select

2. Choose your favourite DNS provider



3. The forwarders will be populated with what you chose

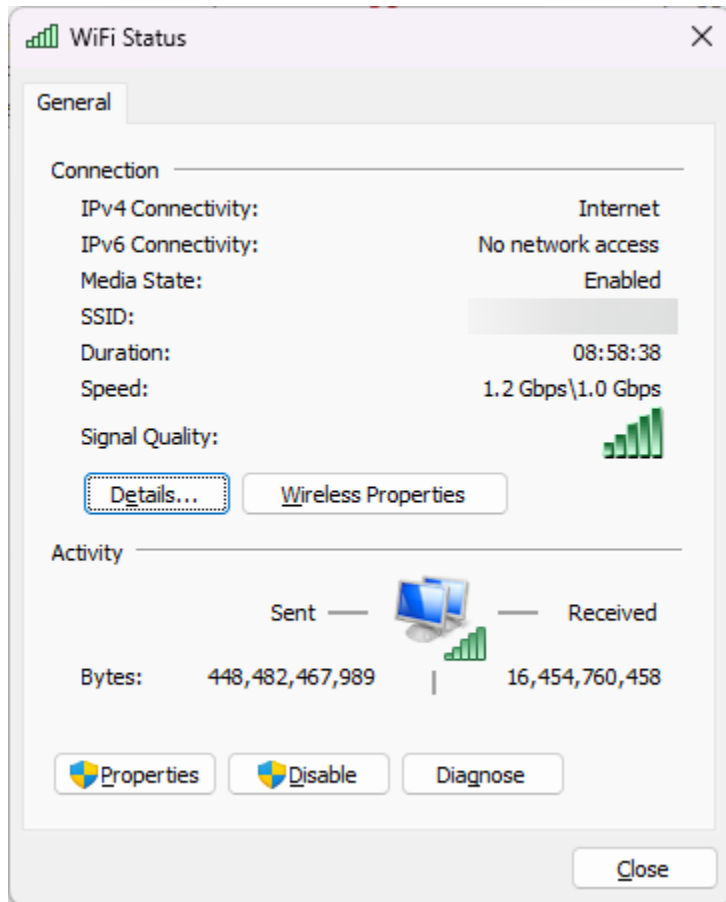


4. Go to the bottom of the page and click on "Save Settings"

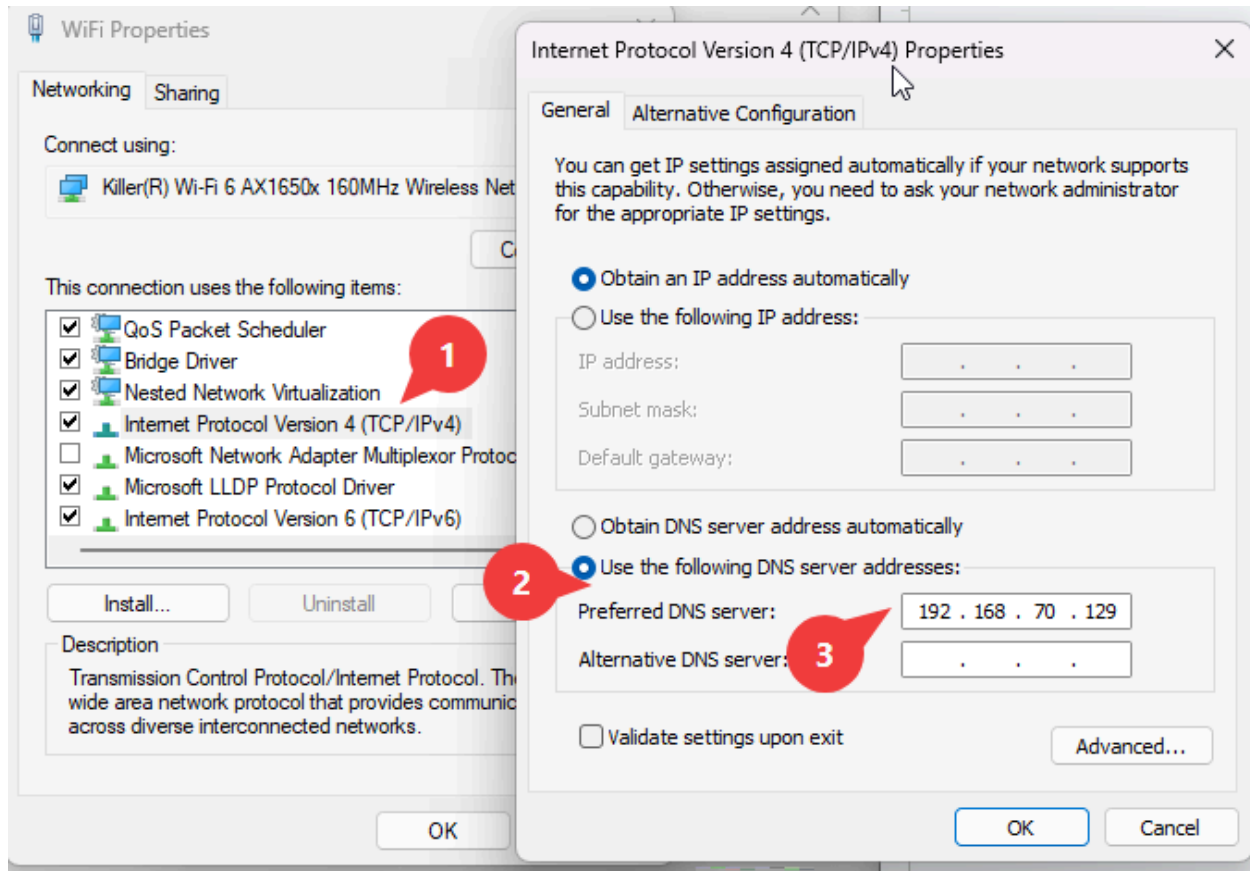
2. Configure your default DNS server

For Windows user only:

1. Win+R to open Run window, type “ncpa.cpl” (which stands for Network control panel applet)
2. Double click on the adapter you are using to connect to the Internet right now (not VMware * Adapter).



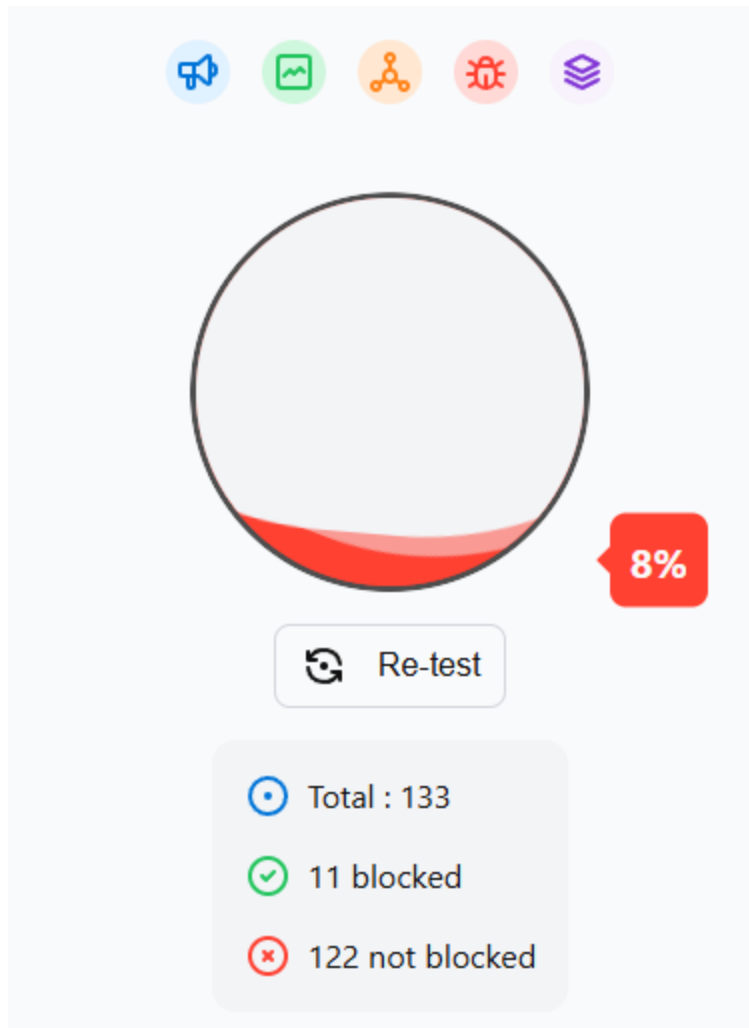
3. Click on Properties



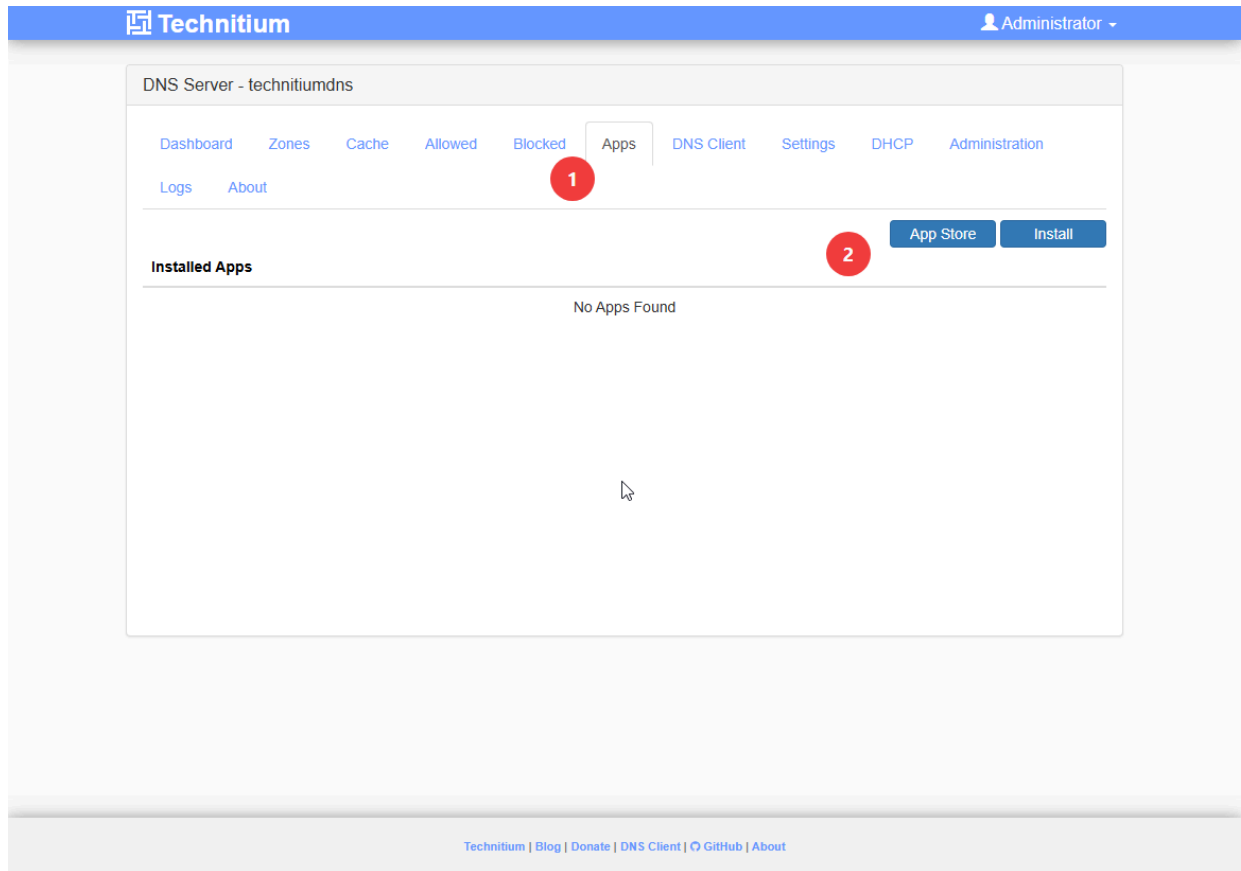
4. Open Internet Protocol Version 4 (TCP/IPv4). Select "Use the following DNS server addresses" and key in the DNS server that we used earlier.
5. Open Command Prompt and type this command `ipconfig /flushdns`.

3. Testing ad-block functionality

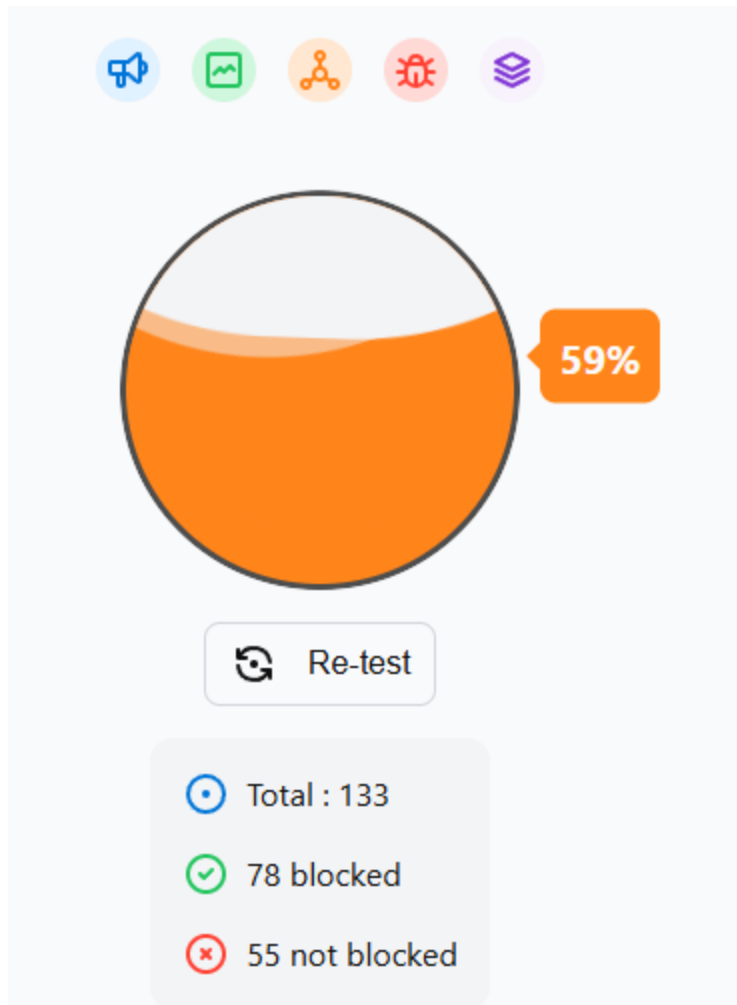
1. Test on <https://adblock.turtlecute.org/> . Make sure before you test it, you should disable your browser's adblocker first.



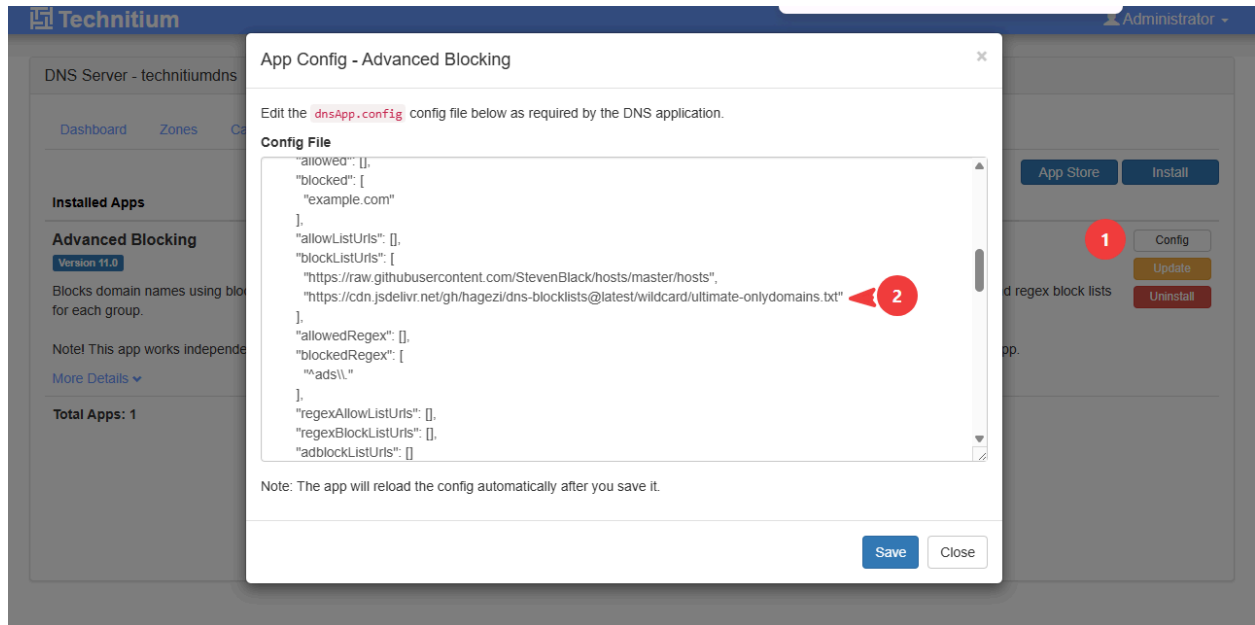
2. Once that is confirmed, let's enable our ad blocker on our DNS server.



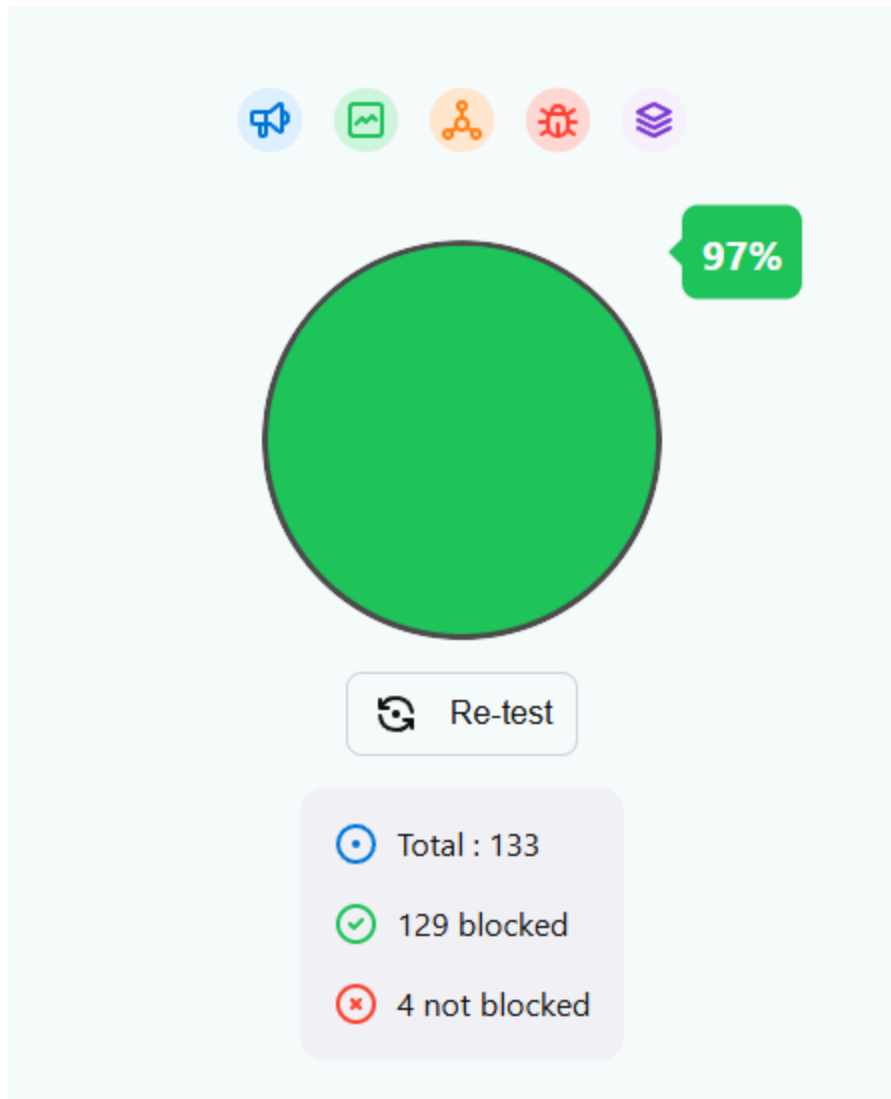
3. Go to Apps and App Store
4. You may go ahead and search around to see what apps are available to install in Technitium DNS. For now, let's install "Advanced Blocking"
 - a. This app provides a block list.



5. We are now able to block 59% as compared to 8% earlier
6. Let's look for more DNS block lists for ads here: <https://github.com/hagezi/dns-blocklists>
7. Make sure to look for "Technitium DNS" under the column "Should be used for". Just for testing purposes, we will be using the ULTIMATE blocker:
<https://cdn.jsdelivr.net/gh/hagezi/dns-blocklists@latest/wildcard/ultimate-onlydomains.txt>
8. Go back to our App in Technitium, and click config
9. Add that link to our "blockListUrls" belonging to the group "everyone else"



10. Test our turtlecute adblocker again



11. This is good enough for me, but if you are interested why it failed, have a look below

👁️ Cosmetic Filter

 Static Ad

 Dynamic Ad

▼ Why does the Cosmetic Filter test fail?

If a cosmetic filter test fails, it simply means that the specific website being tested (in this case `adblock.turtlecute.org`) isn't included in any of adblock's rules or lists.

It's important to note that this does not mean that cosmetic filtering fails on all websites.

You can confirm this by visiting a popular, ad-rich site where you're unlikely to see any ad boxes.

The purpose of this test is to assess the functionality of the blocking feature, not to determine its scope or coverage. By adding the following rules to your adblock solution, you may be able to solve the problem:

```
adblock.turtlecute.org##.adbox.banner_ads.adsbox
adblock.turtlecute.org##.textads
```

Adding these rules could lead to a successful test result proving your adblock solution have that feature of blocking with cosmetic filters

📄 Ad Scripts Loading

`ads.js`

`pagead.js`

▼ Why does the Ad Script Loading test fail?

Same as the cosmetic tests. If an ad script load test fails, it usually means that the specific website being tested isn't covered by any of adblock's rules or lists, especially for blocking ad-related scripts like my fake `ads.js`.

However, this error doesn't indicate a general failure of ad script blocking on all websites.

To check, you can visit a popular website known for its abundance of ads scripts.

Chances are that you won't encounter any blocked ad scripts. It's important to understand that this test is designed to evaluate the functionality of ad script blocking, not its scope or effectiveness. To potentially fix the problem, consider adding the following rules to your adblock solution:

```
/pagead.js$domain=adblock.turtlecute.org
/widget/ads.
```

Adding these rules could lead to a successful test result proving your adblock solution have that feature of blocking script loading

12. "But why does my Youtube still have ads?!" – Youtube no longer use subdomain for ads. To block ads on Youtube using a DNS blocker meant that you must block Youtube altogether.

<https://docs.google.com/document/d/1nqZww1b9Avnp4lokf4P-qWmHcGVMtofD5oILxfVmNkk/edit?usp=sharing>